Classification Dataset Description

Machine Learning | Exercise 0 | Group 56

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We selected MembershipWoes (OpenML 44225, version 2) and Dermatology (UCI 33) for their contrasting sizes, measurement types, and classification tasks. Both have missing inputs and require preprocessing [1, 2].

MembershipWoes has 10,362 records and 14 non-target columns: five quantities, six nominal fields, two dates, and one identifier. Dermatology has 366 patients and 34 inputs.

MembershipWoes

The binary target has 7,219 INFORCE records (69.67%) and 3,143 CANCELLED records (30.33%). Age at issue ranges from 0 to 92 years (median 46); additional members from 0 to 3. Fees span 0–10,100,000 (median 118,209.28), and observed income 9,996–999,999,996. Currency is unspecified. Four zero ages and terms up to 102 years warrant clarification. Occupation and agent codes are nominal; agent code has 4,317 levels.

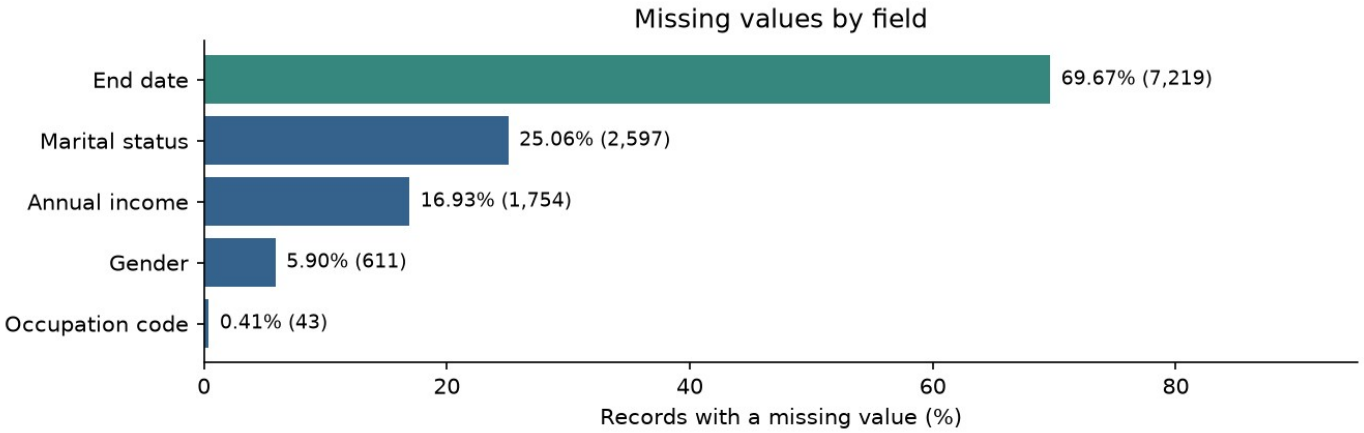


Figure 1. Missing values among all 10,362 records; labels give percentages and counts.

Missingness affects 8,024 rows (77.44%). Every cancelled membership has an end date; every in-force membership lacks one. Exclude end date and its availability indicator unless prediction timing justifies their use. Removing the unique identifier and end date leaves 12 candidate fields.

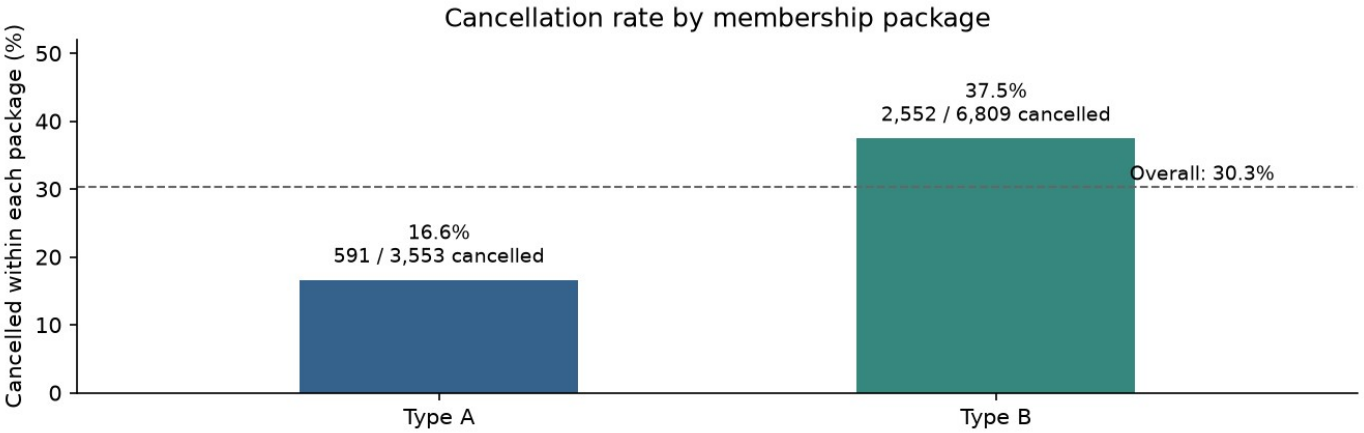


Figure 2. Cancellation percentages within each package, with cancelled/total counts.

Package rates differ: 16.63% for Type A and 37.48% for Type B. Other group differences could contribute. Future modelling requires suitable category encoding, missing-value handling, and checked transformations or scaling of monetary inputs.

Dermatology

The 34 inputs comprise 12 clinical variables and 22 biopsy-based findings. Thirty-two scores have ordinal degrees 0–3 (absent to largest); equal clinical intervals are not established. Family history is binary nominal, age quantitative, and disease codes nominal [2].

Class counts are psoriasis 112 (30.60%), seborrhoeic dermatitis 61 (16.67%), lichen planus 72 (19.67%), pityriasis rosea 49 (13.39%), chronic dermatitis 52 (14.21%), and pityriasis rubra pilaris (PRP) 20 (5.46%). The 5.6-fold largest/smallest imbalance supports stratified evaluation and per-class metrics.

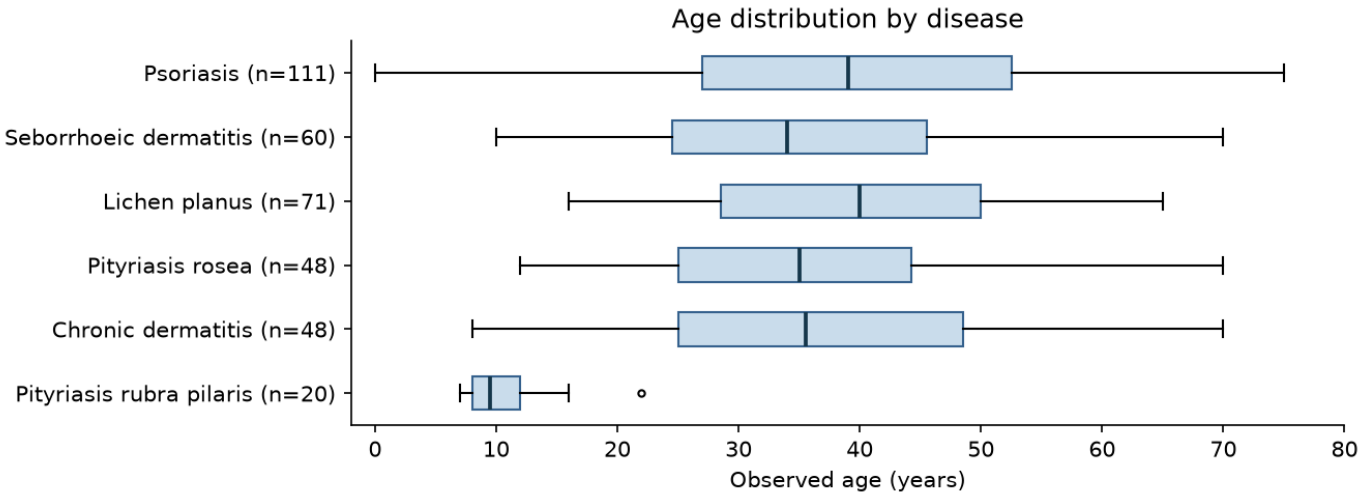


Figure 3. Observed ages by disease. Boxes show the middle 50% and medians; whiskers use 1.5 × IQR. Eight missing ages are omitted; zero age is retained.

Age is the only incomplete input (8/366, 2.19%). Observed ages span 0–75 (median 35). PRP has median 9.5 years, compared with 34–40 for the other classes. These sample patterns do not establish diagnostic rules.

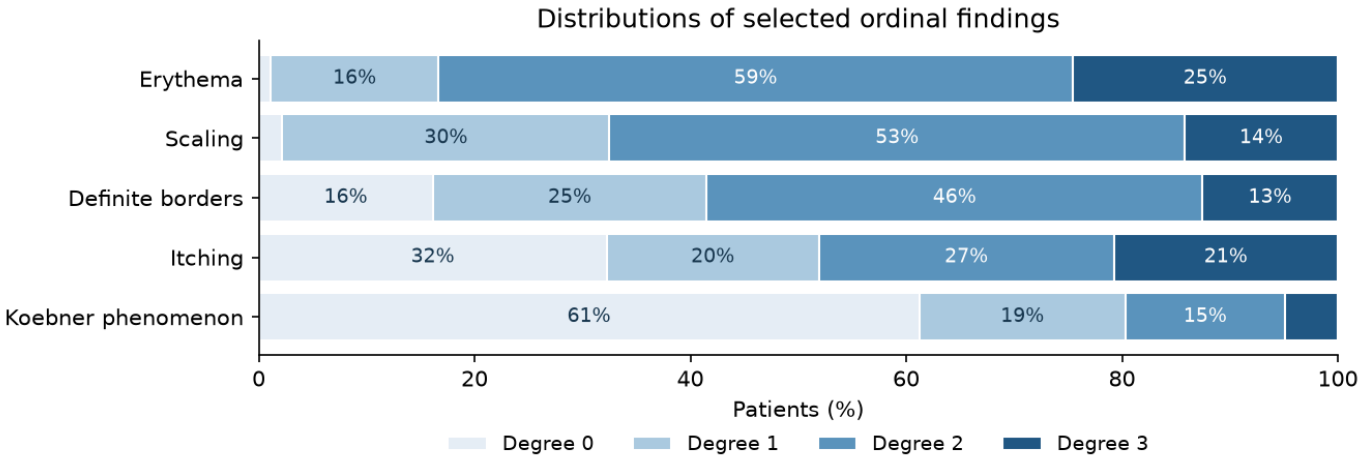


Figure 4. First five clinical scores in documented order; each bar includes all 366 patients. Segment labels show rounded percentages of at least 8%.

Scaling degree 2 is most frequent (53.28%); family history is positive in 46 patients (12.57%). No complete duplicate rows occur. Preserve score order and fit any age imputation or scaling only on training folds. Biopsy-dependent inputs limit interpretation as a clinical-only screening task.

Sources and reproducibility

[1] OpenML. MembershipWoes, dataset 44225, version 2. https://www.openml.org/search?type=data&status=active&id=44225
[2] Ilter, N., & Guvenir, H. (1998). Dermatology [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5FK5P
Accessed 24 October 2026. The accompanying notebooks record source fingerprints and package versions, and contain all analysis and plotting code. Figures are copied directly from executed notebook outputs.